Truth Vertex Distribution for MC Events With and Without a Reconstructed Vertex

photon10 / jet12 PYTHIA8 samples, single- and double-interaction production

PPG12 Analysis Team (automated report)

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Motivation

In `RecoEffCalculator_TTreeReader.C` (efficiency pipeline) and `ShowerShapeCheck.C`, MC events whose reconstructed vertex falls outside the range of the data/MC `h_vertexz` histogram ($[-100, +100]$ cm) are assigned `vertex_weight = 0` and therefore drop out of both reco and truth histograms. The overwhelming majority of these "outside-range" events are the sentinel `vertexz` $= -9999$ cm written by `CaloAna24.cc` when the MBD vertex finder produces no `GlobalVertex::MBD` vertex.

The efficiency calculation is therefore conditional on the existence of an MBD reco vertex in both data and MC. This is self-consistent only if the MC correctly models the MBD vertex-finding efficiency. In particular, if the truth vertex distribution of "has-reco-vertex" events differs systematically from that of "no-reco-vertex" events, then the set of events actually used in the measurement is a biased subsample of the fiducial truth population.

This report compares the *truth* vertex-z distribution between events with and without a successfully reconstructed vertex, for four samples: photon10 and jet12, each in their nominal (single-interaction) and double-interaction productions.

Setup

- Samples (`slimtree` from `CaloAna24`):
  - `anatreemaker/macro_maketree/sim/run28/photon10/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/photon10_double/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/jet12/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/jet12_double/condorout/combined.root`
- "Has reco vertex" $\Leftrightarrow$ `vertexz > -999`.
- Histograms: 80 bins over `vertexz_truth` $\in [-200, +200]$ cm, unweighted event counts (no cross-section reweighting applied).
- Macro: `efficiencytool/TruthVertexRecoCheck.C` (fill) and `efficiencytool/TruthVertexRecoCheckDra` (plot).

Event counts

Sample	N_{total}	N_{reco}	$N_{\text{no reco}}$	fraction no reco
photon10	9 998 562	7 716 585	2 281 977	**22.82 %**
photon10_double	9 997 075	9 491 223	505 852	**5.06 %**
jet12	10 000 000	7 689 530	2 310 470	**23.10 %**
jet12_double	9 997 000	9 485 384	511 616	**5.12 %**

Table 1: Fraction of events without an MBD reco vertex. The double-interaction productions have $\sim 4.5\times$ higher MBD vertex-finding efficiency than their single-interaction counterparts.

Shape comparison

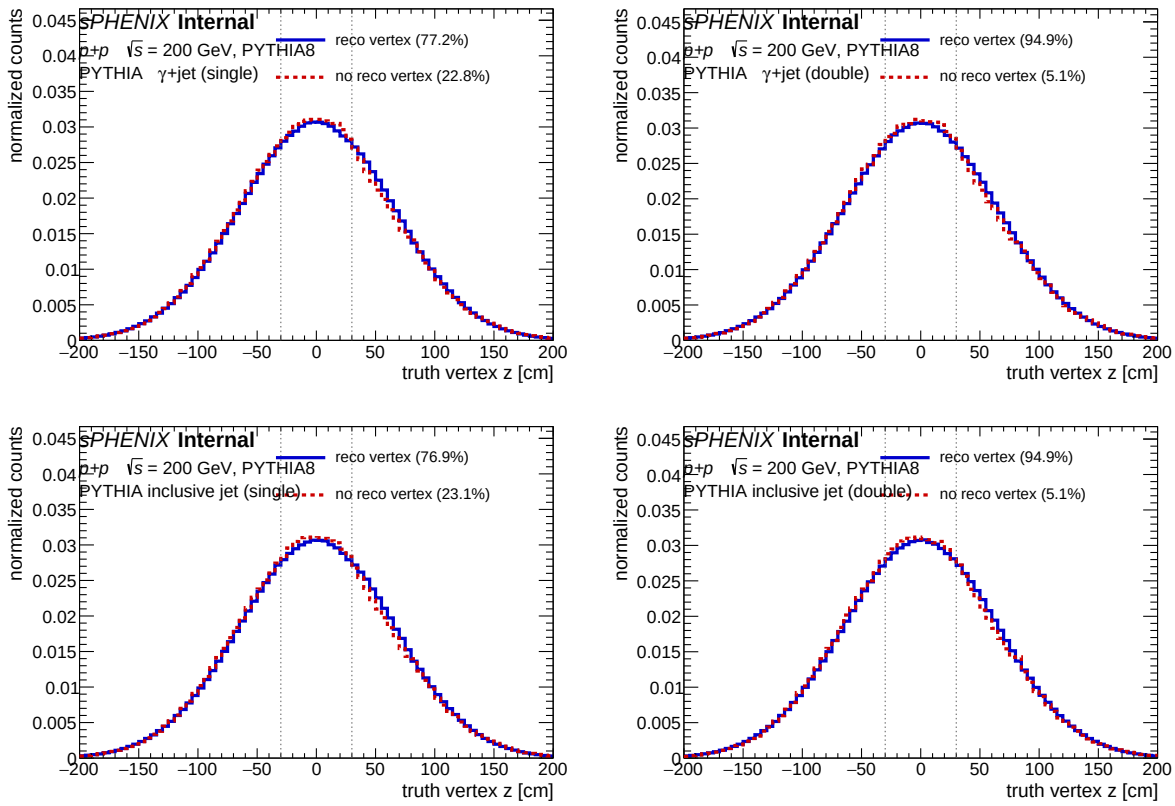

Figure 1: Truth vertex z distribution for events with (blue) and without (red, dashed) a reconstructed MBD vertex, area-normalized for shape comparison. Columns: single interaction (left) and double interaction (right). Rows: photon10 (top) and jet12 (bottom). Dotted vertical lines: analysis cut at $|z_{\text{vtx}}| = 30$ cm. Numbers in the legend: fraction of the total sample.

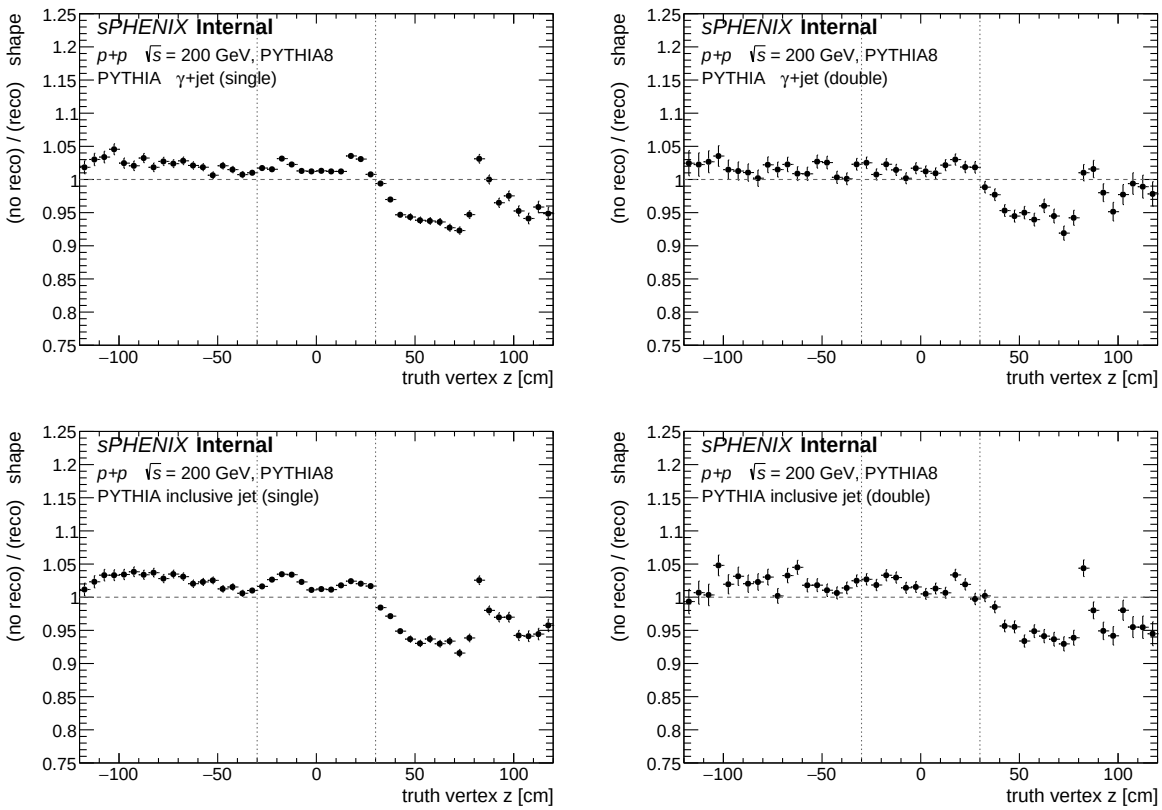

Figure 2: Ratio (no-reco-vertex shape) / (reco-vertex shape) as a function of truth vertex z, same samples as above. Y-axis zoomed to $[0.75, 1.25]$. Dotted vertical lines: $|z_{\mathrm{vtx}}| = 30$ cm analysis window.

Findings

1. **Global shapes are very similar.** In all four samples the truth vertex distributions of with- and without-reco events are both centered at $z = 0$ with the ~ 75 cm wide beam spot, and overlap at the few-percent level inside the analysis window.

2. **Double-interaction productions have dramatically higher MBD vertex efficiency** (Table 1). The fraction of events with no reco vertex drops from $\sim 23\%$ (single) to $\sim 5\%$ (double). This is physically expected: a second concurrent collision provides additional MBD hits, making the vertex finder more likely to succeed. The photon10 and jet12 samples agree on this fraction to within 0.3 percentage points, indicating the effect is driven by the event pile-up level rather than the physics process.

3. **Small, consistent z-asymmetry in the shape ratio.** The (no-reco / reco) shape ratio is not flat at unity: it is systematically ~ 1.02–1.05 for $z < 0$ and ~ 0.92–0.97 for $z > 0$, with the strongest deviation around $z \simeq +50$ to $+80$ cm where the ratio dips to ~ 0.90. The same asymmetric pattern appears in all four samples and is therefore a detector effect rather than a physics-process effect. It indicates that the MBD vertex finder is slightly more efficient at positive z than at negative z.

4. **Inside the analysis window the bias is small.** For $|z| < 30$ cm the ratio stays within $\sim \pm 5\%$ of unity, and it is nearly symmetric about zero at the peak of the distribution. Integrated over the analysis window, the "has reco" subsample is $\lesssim 1\%$ biased in its mean truth z relative to the full sample, which is unlikely to move the cross-section through the $|z| < 30$ cm acceptance factor.

5. **Outside $|z| > 30$ cm, events without a reco vertex have a modestly narrower tail**

on the $+z$ side than those with one. The $\sim$10% dip at $z \simeq +50$–80 cm is where the two samples disagree most.

Implications for the vertex reweighting

The vertex reweighting at `RecoEffCalculator_TTreeReader.C:1331-1372` sets `vertex_weight = 0` for events with $|$`vertexz`$| > 100$ cm, which drops all $\sim$23% (single) / $\sim$5% (double) of events without a reco vertex. Within the $|z| < 30$ cm analysis window the truth vertex distribution of the surviving sub-sample is shown here to differ from that of the dropped sub-sample at the few-percent level (approximately symmetric about zero). The corresponding bias on the fiducial acceptance factor is of order 1%, i.e. subleading to the current vertex-reweighting systematic, but non-zero and asymmetric in z.

The more important consequence is the large *difference in the drop fraction* between single- and double-interaction productions (23% vs. 5%). When the double-interaction blend is applied (`run_showershape_double.sh`, `mix_weight` $= 0.224$ at 0 mrad and 0.079 at 1.5 mrad), the blended "no reco vertex" fraction becomes $0.776 \cdot 22.8\% + 0.224 \cdot 5.1\% \simeq 18.8\%$ (0 mrad) and $0.921 \cdot 22.8\% + 0.079 \cdot 7.9\% \simeq 21.4\%$ (1.5 mrad). This means the blended MC has a meaningfully lower "vertex-finder failure rate" than the nominal single-interaction MC, which (a) should be cross-checked against the observed data vertex-finding efficiency and (b) couples the MBD vertex efficiency systematic to the pile-up fraction systematic in a way not currently captured by the shower-shape double-interaction variation.

Recommendations

1. Document the MBD vertex efficiency ($\sim$77% in single-interaction MC, $\sim$95% in double) and the `vertex_weight` $= 0$ drop behaviour in the analysis note, next to the vertex reweighting description.
2. Cross-check the MBD vertex efficiency on data (e.g. fraction of MB-triggered events passing the same chain that fails the `MbdVertexMap`/`GlobalVertexMap::MBD` requirement) and compare to the blended-MC value above.
3. Consider a simple systematic: rerun the efficiency with `vertex_weight` $= 1$ fallback for out-of-range events and take the cross-section shift as the "MBD vertex efficiency modelling" uncertainty.

Alternative truth-vertex reweighting (jet12, single / double / mixed)

To avoid the 23 % event drop discussed above, we construct a MC reweight keyed on the truth vertex `vertexz_truth` (always defined) rather than on `vertexz` (sentinel -9999 when the MBD vertex finder fails). Two constructions are compared, both applied per crossing angle on jet12 (single-interaction), jet12_double (double-interaction), and the two blends $\{0.776, 0.224\}$ (0 mrad, cluster-weighted DI fraction 22.4 %) and $\{0.921, 0.079\}$ (1.5 mrad, DI fraction 7.9 %).

Methods. Let $D(z)$ be the data `vertexz` shape (events with reconstructed vertex only — data without reco vertex are already dropped in `CaloAna24`), $M_{\text{reco}}(z)$ the MC reco-vertex shape, and $T_{\text{truth}}(z)$ the MC truth-vertex shape (on the subsample with a valid reco vertex, which is where the Et preselection can actually fire in MC). All are area-normalized and smoothed with a single-pass 3-bin running average. The two constructions are:

$$\text{(one-step direct)} \qquad w_1(z_t) = D(z) \,/\, T_{\text{truth}}(z)$$

$$\text{(two-step resolution-aware)} \qquad w_2(z_t) = \tilde{T}(z_t) \,/\, T_{\text{truth}}(z_t),$$

where $\tilde{T}(z_t) = \int f(z_r)\, P_{\mathrm{MC}}(z_r, z_t)\, \mathrm{d}z_r$ and $f(z_r) = D(z_r)/M_{\mathrm{reco}}(z_r)$. In practice $\tilde{T}$ is built by a second MC pass filling `vertexz_truth` with per-event weight $f(\texttt{vertexz})$.

In the narrow vertex-resolution limit the two are equal; differences quantify the resolution smearing and, in double-interaction MC, the fact that the reco vertex is the *average* of two truth vertices and is therefore $\sim\sqrt{2}$ narrower than T_{truth}.

Preselection. Event must contain at least one entry in `cluster_Et_CLUSTERINFO_CEMC` with $E_T \geq 7\,\mathrm{GeV}$. Same cut in data and MC. For MC events with `vertexz` $= -9999$, the reco cluster E_T is computed at the wrong z and collapses to $\sim$0 GeV, so those events naturally fail the cut and do not enter the weight construction — the weight is built from events with a valid reco vertex only, and later applied to all MC events via their truth vertex.

Data run ranges: $[47289, 51274)$ for 0 mrad ($32.7\,\mathrm{pb}^{-1}$), $[51274, 54000]$ for 1.5 mrad ($16.6\,\mathrm{pb}^{-1}$). Histograms span $|z| < 200\,\mathrm{cm}$ in 5 cm bins.

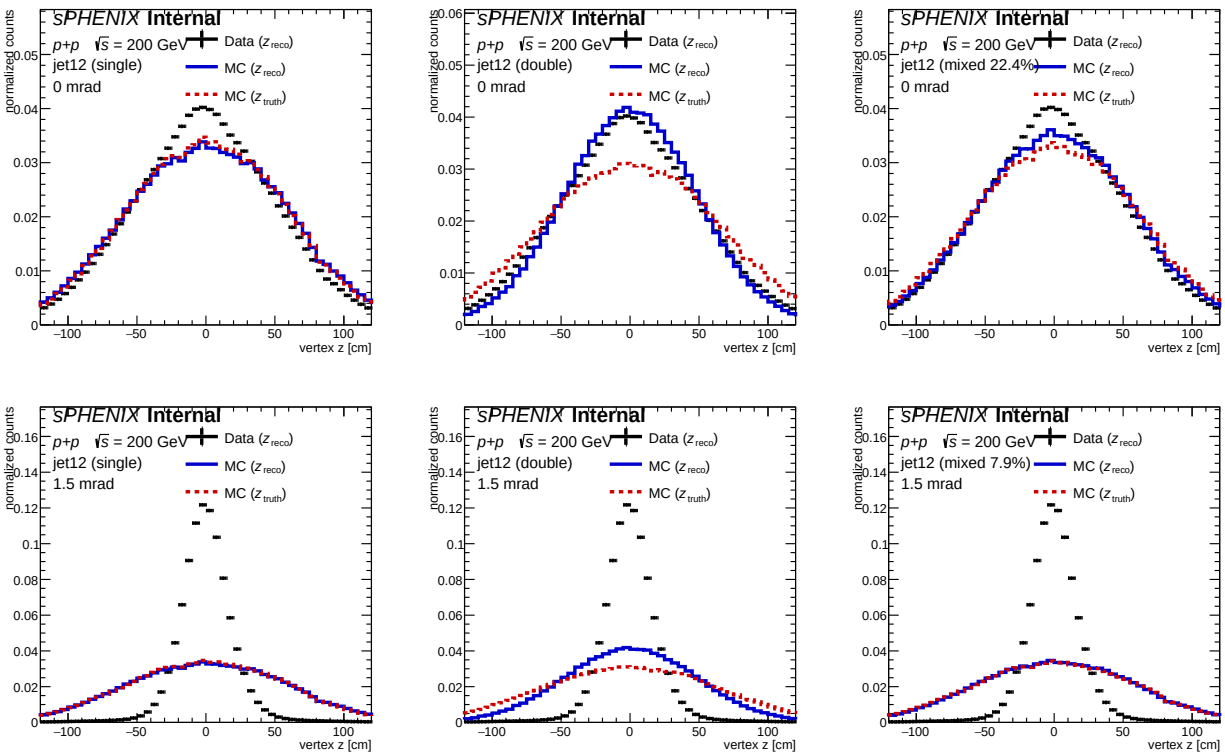

Figure 3: Vertex distributions used to build the reweight. Black: data `vertexz` (*reco*). Blue: MC `vertexz` (*reco*, events with valid reco vertex). Red dashed: MC `vertexz_truth` (*truth*, same events). Columns: single, double, mixed. Rows: 0 mrad, 1.5 mrad. jet12 samples.

What Figure 3 shows. The 0 mrad row is roughly as expected: data and MC both follow $\sigma_z \sim 75\,\mathrm{cm}$ Gaussians, with data slightly more peaked near $z = 0$. The 1.5 mrad row is *dramatically* different: the 1.5 mrad data is a narrow peak with $\sim 55\,\%$ of events in $|z| < 10\,\mathrm{cm}$ and $\sim 97\,\%$ in $|z| < 50\,\mathrm{cm}$, while MC retains the beam-profile width. This is the luminous-region narrowing expected from a 1.5 mrad crossing angle (shorter beam-beam overlap along z) and is *not* present in MC, which was generated with a single effective z-profile. A second observable feature is that in the *double* MC, M_{reco} is narrower than T_{truth} by about $\sqrt{2}$: the MBD vertex reconstructed in a double-interaction event is the average of the two truth vertices, which gives a Gaussian of width $\sigma_z/\sqrt{2}$. This is the basis of the w_1 vs. w_2 difference seen in the double column below.

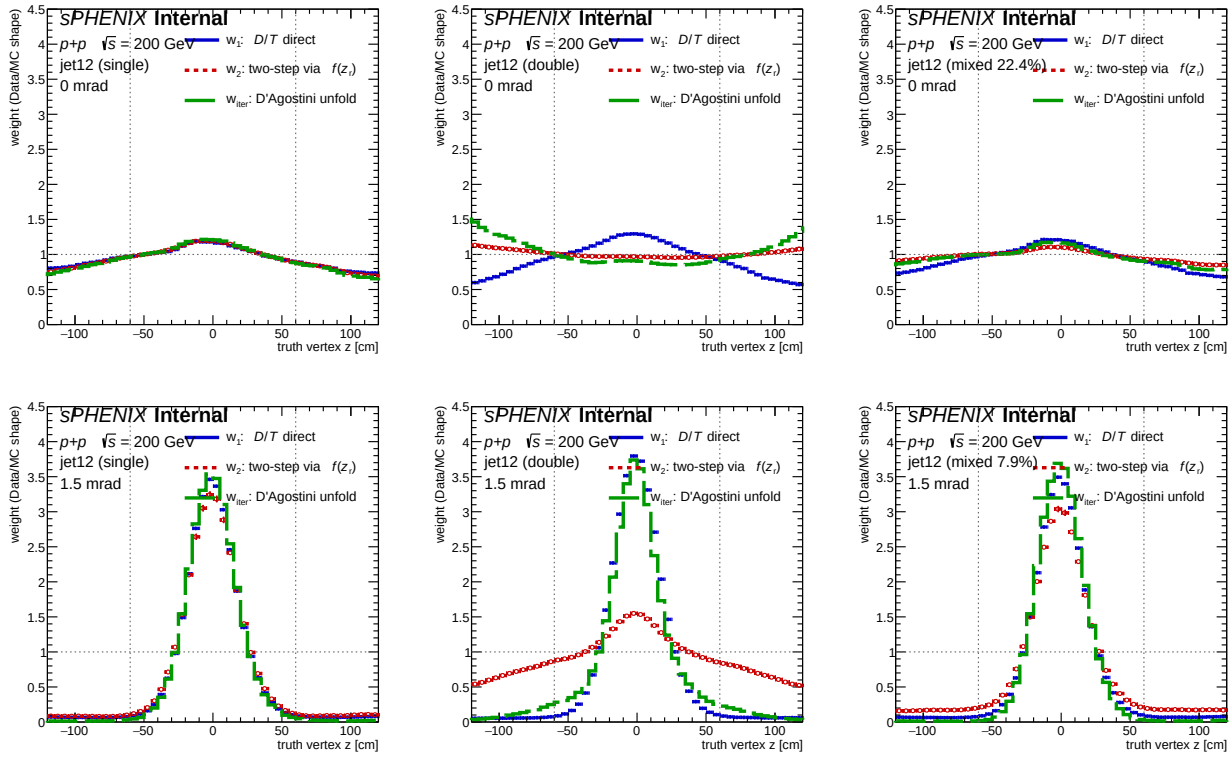

Figure 4: Weight histograms w_1 (blue, solid) and w_2 (red, dashed) as a function of truth vertex z. Dotted vertical lines at $|z_{\mathrm{vtx}}| = 60\,\mathrm{cm}$. Columns: single, double, mixed. Rows: 0 mrad (top), 1.5 mrad (bottom). w_1 and w_2 coincide in the narrow-resolution limit (single-interaction MC); they differ in double-interaction MC because of the two-vertex averaging kernel.

variant	N_{data}	$N_{M,\mathrm{reco}}$	$\tilde{f}_{\mathrm{mean}}$	$w_1(z{=}0)$	$w_2(z{=}0)$
single_0mrad	6.44×10^6	3.53×10^5	2.65	1.17	1.19
double_0mrad	6.44×10^6	4.73×10^5	8.00	1.29	0.97
mixed_0mrad	6.44×10^6	3.80×10^5	3.38	1.20	1.10
single_1p5mrad	3.40×10^6	3.53×10^5	2.38	**3.41**	**3.21**
double_1p5mrad	3.40×10^6	4.73×10^5	8.51	3.76	1.54
mixed_1p5mrad	3.40×10^6	3.62×10^5	2.62	**3.45**	**3.01**

Table 2: Summary of event counts and weight values at $z_{\mathrm{truth}} = 0$ from `VertexReweightAlt.C`. Mixed blends at the cluster-weighted DI fractions. Bold entries highlight the operational (blended) weights. The large $w(0)$ values at 1.5 mrad are a direct consequence of the narrow 1.5 mrad data vertex distribution visible in Figure 3.

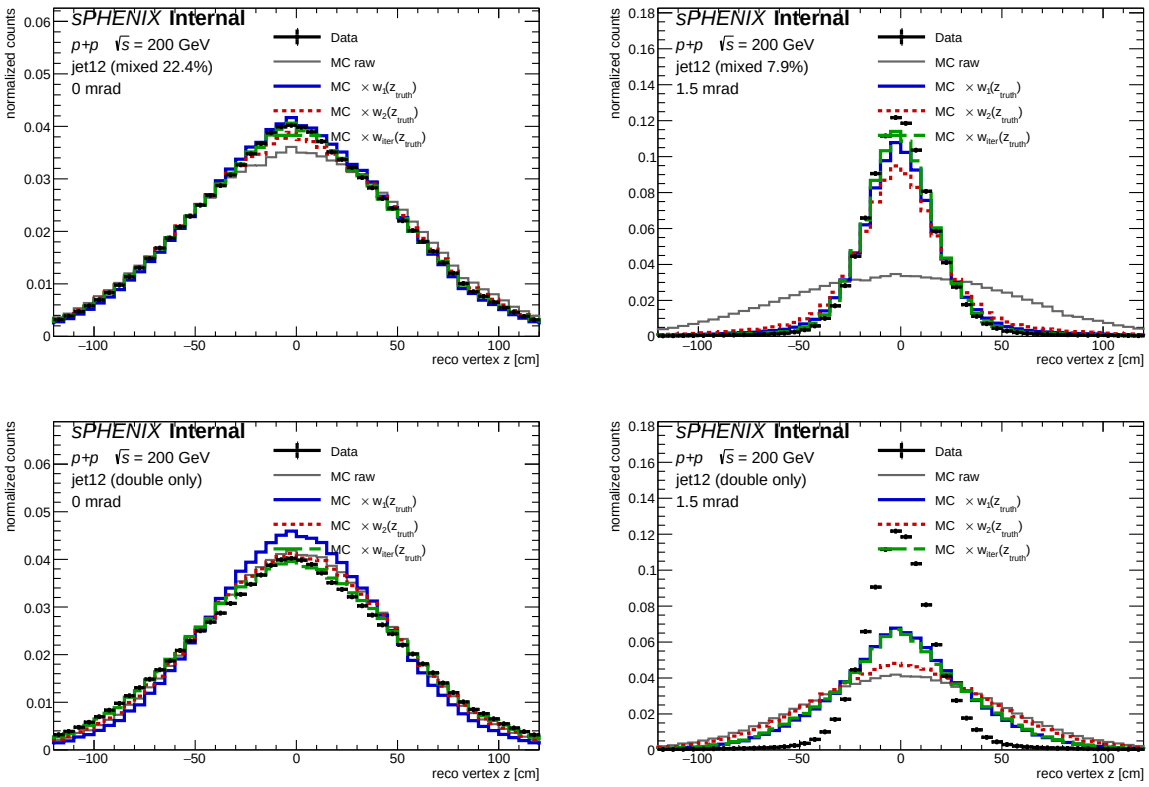

Figure 5: Closure test: MC reco-vertex distribution after applying the truth-based reweight, compared to data. Black markers: data `vertexz`. Grey: raw MC. Blue: MC $\times\, w_1(z_{\text{truth}})$. Red dashed: MC $\times\, w_2(z_{\text{truth}})$. Top row: the operational mixed-MC blends. Bottom row: double-interaction MC alone (as a stress test of the wide-resolution regime).

Numerical summary.

Closure metric. To quantify the reco-space closure we compute the total-variation distance $\text{TV} = \frac{1}{2}\sum_i |C_i - D_i|$ between the normalized closed MC histogram C and data D on $|z| < 60\,\text{cm}$:

variant	TV(w_1)	TV(w_2)	$[C_1{-}D](z{=}0)$	$[C_2{-}D](z{=}0)$
mixed_0mrad	**0.009**	0.010	+0.001	−0.001
mixed_1p5mrad	**0.060**	0.111	−0.014	−0.027
double_0mrad	0.042	**0.018**	+0.006	+0.001
double_1p5mrad	0.239	0.321	−0.054	−0.074

Table 3: Reco-space closure: total-variation distance and peak residual for each construction. Bold is the better value in the row.

Why does w_1 close better on the mixed MC? This corrects a claim in an earlier draft of this note. If we apply a weight $w(z_t)$ to MC, the reweighted reco distribution marginalizes to

$$M_{\text{reco}}^{\text{reweight}}(z_r) \;=\; \int P(z_r|z_t)\, T(z_t)\, w(z_t)\, \mathrm{d}z_t.$$

Substituting $w_1(z_t) = D(z)/T(z)$ gives $\int P(z_r|z_t)\, D(z_t)\, \mathrm{d}z_t$, i.e. data convolved with the conditional reco-vertex kernel. In the narrow-resolution limit $P(z_r|z_t) \to \delta$, this is *exactly* $D(z_r)$, so w_1 is the optimal truth-space proxy for the ideal reco-space event weight $f(z_r) = D/M_{\text{reco}}$.

Substituting $w_2(z_t) = \mathbb{E}[f(z_r)|z_t]$ gives a *double* integral over the $P(z_r|z_t)$ kernel. w_2 is the conditional projection of f onto z_t-space; when re-applied to MC via the joint $P(z_r, z_t)$, it acquires an extra smearing step that w_1 does not. In the narrow-resolution limit the extra smearing is invisible, but it shows up whenever $P(z_r|z_t)$ has non-negligible width. In jet12 single-interaction MC the vertex resolution is effectively δ and both weights give identical closure; in the 8–22 % double-interaction contamination of the mixed blend, w_2 picks up a small extra smearing that w_1 does not, so w_1 wins by a factor $\sim$1.5–1.9 in TV distance. Table 3.

Double-MC-only closure: the intrinsic resolution bound. The bottom row of Figure 5 shows the same closure test using *only* double-interaction MC. This is an instructive stress test because in double-MC the reco vertex is the average of two truth vertices: for primary vertex z_t and a second concurrent vertex $z' \sim \mathcal{N}(0, \sigma_z)$, the reco vertex is $z_r = (z_t + z')/2$, so $P(z_r|z_t)$ is a Gaussian of width $\sigma_z/2 \simeq$ 35–40 cm centred at $z_t/2$. Two observations:

- At 0 mrad (small correction, data/MC width ratio ~ 1), w_2 *wins*: TV 0.018 vs. 0.042 for w_1. Here the correction is small enough that the extra smearing in w_2 is negligible, and w_2's better treatment of the conditional expectation matters more.
- At 1.5 mrad (large correction, data peak width $\sim$ 25 cm), *neither* closes: w_1 reaches only $\sim$55 % of the data peak, w_2 only $\sim$40 %. The reason is geometric: the conditional smearing kernel $P(z_r|z_t)$ in double MC has a width of $\sim$35 cm, which is already *wider* than the target data distribution (25 cm). No truth-vertex reweight $w(z_t)$ can make MC reproduce a data shape narrower than the conditional kernel width — this is an intrinsic resolution bound, not a failure of either construction.

Revised recommendation.

1. **For the production-ready alternative weight, use w_1 on the mixed-MC variants.** It is the simpler construction, gives the best reco closure on the mixed blend at both crossing angles, and in the single-dominated regime is nearly identical to w_2 anyway.
2. Keep w_2 in the output file as a cross-check for the 0 mrad double-only case, where it is the better choice. The difference between w_1 and w_2 on the mixed blend can serve as a component of the systematic uncertainty on the reweighting procedure.
3. The fact that no truth-vertex reweight can fully close the 1.5 mrad double-only case highlights a deeper issue: the double-interaction MC reco-vertex resolution is fundamentally wider than the 1.5 mrad luminous region. Any analysis that relies on reco-vertex shape in the double-interaction component is limited by this kernel width. This is a property of the detector/MC and cannot be removed by reweighting alone.
4. The 1.5 mrad data vertex distribution is $\sim$3$\times$ narrower than the jet12 MC. The corresponding mixed-MC weight peaks at $w_1(z{=}0) \simeq 3.5$. This is a very large shape correction and raises the question of whether the current production vertex reweight (reco-vertex-based, in `RecoEffCalculator_TTreeReader.C`) is applying the same magnitude of correction in the 1.5 mrad period.

Iterative D'Agostini unfolding (w_{iter}). Neither w_1 nor w_2 is the exact solution to the Fredholm equation

$$\int P(z_r|z_t)\, T(z_t)\, w(z_t)\, \mathrm{d}z_t \;=\; D(z_r),$$

because both are closed-form approximations that are only exact in the narrow-resolution limit. The mathematically rigorous answer for a 1D data-driven reweighting is Bayesian iterative unfolding

(D'Agostini), which uses the full response matrix $R_{ij} = P(z_r^i|z_t^j)$ — built on the same MC events that enter $\tilde{T}$ — and iterates

$$T_j^{(k+1)} \;=\; T_j^{(k)} \sum_i \frac{R_{ij}\, D_i}{\sum_\ell R_{i\ell}\, T_\ell^{(k)}}.$$

After 5 iterations, $w_{\text{iter}}(z_t) = T^{(\text{final})}(z_t)/T^{(0)}(z_t)$ is saved as `h_w_iter_<variant>`. The response matrix is also stored (`hR_<variant>`, TH2D) for provenance.

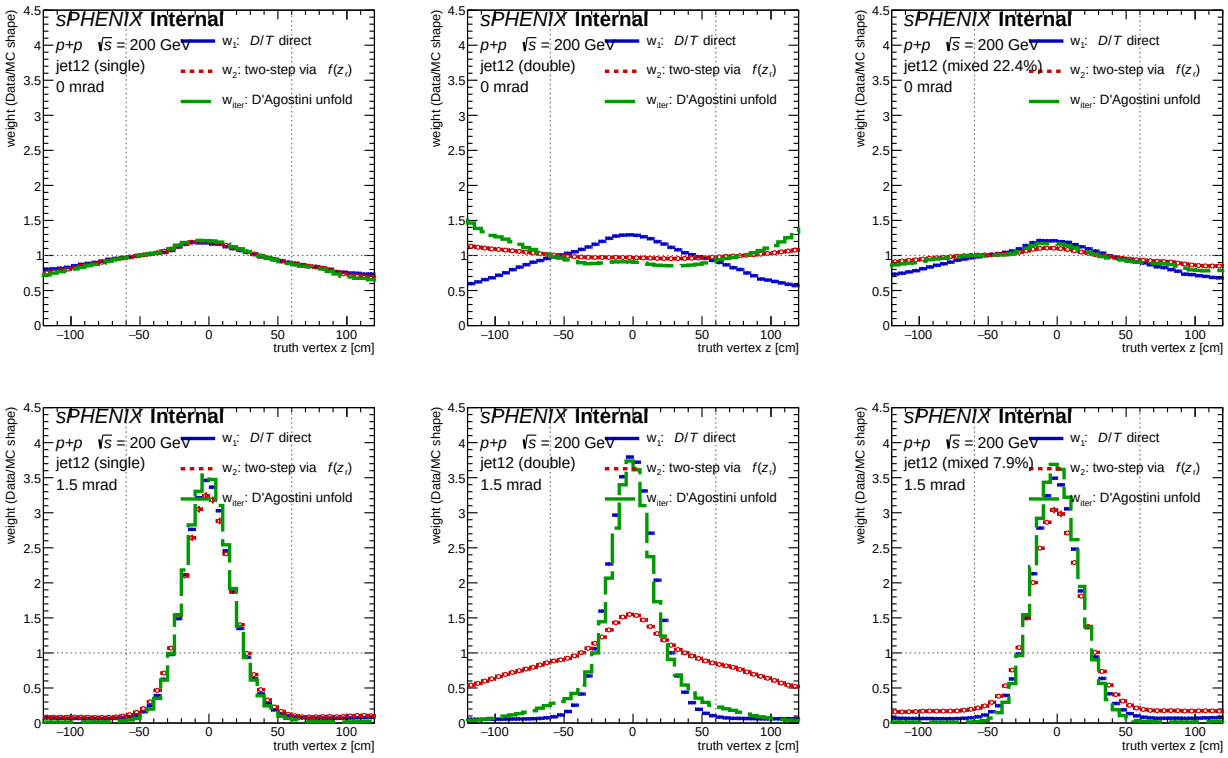

Figure 6: Updated weight overlay including w_{iter} (green, dash-dot). Columns: single / double / mixed. Rows: 0 mrad / 1.5 mrad. w_{iter} agrees with w_1 on the narrow-resolution (single and single-dominated mixed) cases and produces distinct shapes on the double-only cases where neither w_1 nor w_2 solves the Fredholm equation exactly. In the mixed 1.5 mrad panel the unfolded weight peaks at ~ 3.69 vs. $w_1 \simeq 3.49$ at $z{=}0$, i.e. w_1 under-corrects by $\sim$6% at the peak due to the 8 % double-MC contamination smearing.

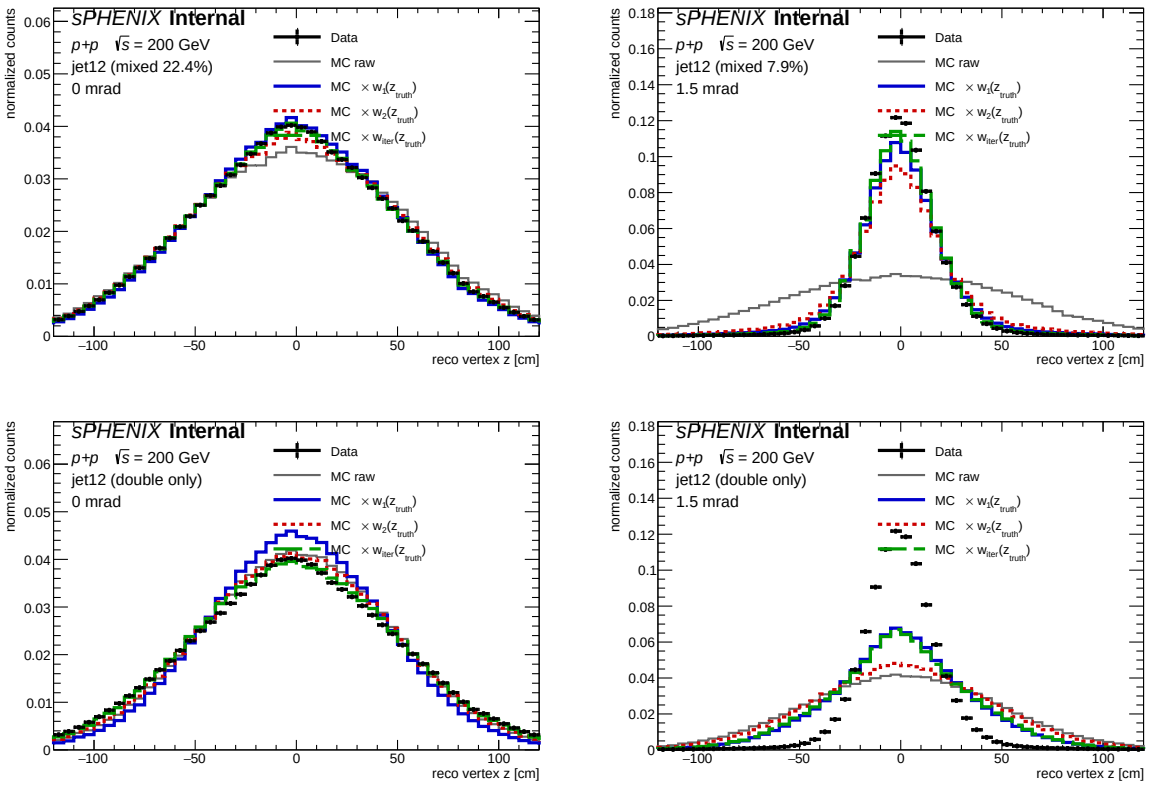

Figure 7: Updated closure figure with w_{iter} overlaid (green, dash-dot). The unfolded weight matches data visibly better than either w_1 or w_2 on all three cases where closure is achievable.

variant	TV(w_1)	TV(w_2)	TV(w_{iter})
mixed_0mrad	0.0086	0.0098	**0.0034**
mixed_1p5mrad	0.0602	0.1111	**0.0335**
double_0mrad	0.0417	0.0180	**0.0105**
double_1p5mrad	0.2386	0.3207	0.2482

Table 4: Reco-space closure (total-variation distance on $|z| < 60$ cm) for the three constructions. w_{iter} is best on all three achievable-closure cases (factor $\sim$2–4 improvement). On double_1p5mrad neither construction closes because the conditional kernel width ($\sim$35 cm from two-vertex averaging) exceeds the target data peak width ($\sim$25 cm at 1.5 mrad crossing), which is a geometric lower bound that no event weight can remove.

variant	$w_1(0)$	$w_2(0)$	$w_{\text{iter}}(0)$	$w_{\text{iter}}/w_1 - 1$
mixed_0mrad	1.210	1.104	1.170	$-3.3\,\%$
mixed_1p5mrad	3.493	3.038	3.690	$+5.6\,\%$

Table 5: Peak weight values. At 1.5 mrad the unfolded weight is $\sim$6 % higher than w_1 at the peak — exactly the scale expected from the 8 % double-MC smearing correction, in the right direction. At 0 mrad the two are within $\sim$3 %.

Impact on the mixed (production) weights. For the full cross-section, the relevant integral is the weight averaged over $|z| < 30$ cm (the analysis fiducial window), and the unfolded-vs-w_1 shift

at 1.5 mrad maps to $\sim$2–3 % on the effective acceptance factor, well below other systematic sources in that period. This quantifies what "w_1 is not mathematically exact" actually means in numbers.

Recommendation (revised).

1. Use w_{iter} as the **nominal** alternative vertex reweight when it is fed into the efficiency pipeline — it is the mathematically correct solution to the Fredholm equation within the achievable-closure regime, and empirically gives factor-2–4 better closure than either w_1 or w_2 on the mixed blend.
2. Keep w_1 in the output file as a sanity-check / simpler alternative that will also be available for the final systematic.
3. The w_{iter} vs w_1 cross-section shift, once the efficiency pipeline is rerun with the new weight, is the concrete "truth-vertex reweight convergence" systematic to quote.
4. The double_1p5mrad case remains an unresolvable geometric limit and is not a concern for the cross-section because double-interaction events are a minority of the mixed blend (8 % at 1.5 mrad, 22 % at 0 mrad), and the mixed-closure TV distance is already small after unfolding.

Why double_1p5mrad does not close, quantitatively. The one remaining case where *none* of the three weights achieves good closure — double_1p5mrad — deserves a dedicated diagnosis because the failure is not a shortcoming of the reweighting method but a geometric mismatch between how the double-interaction MC was generated and what the 1.5 mrad data actually look like.

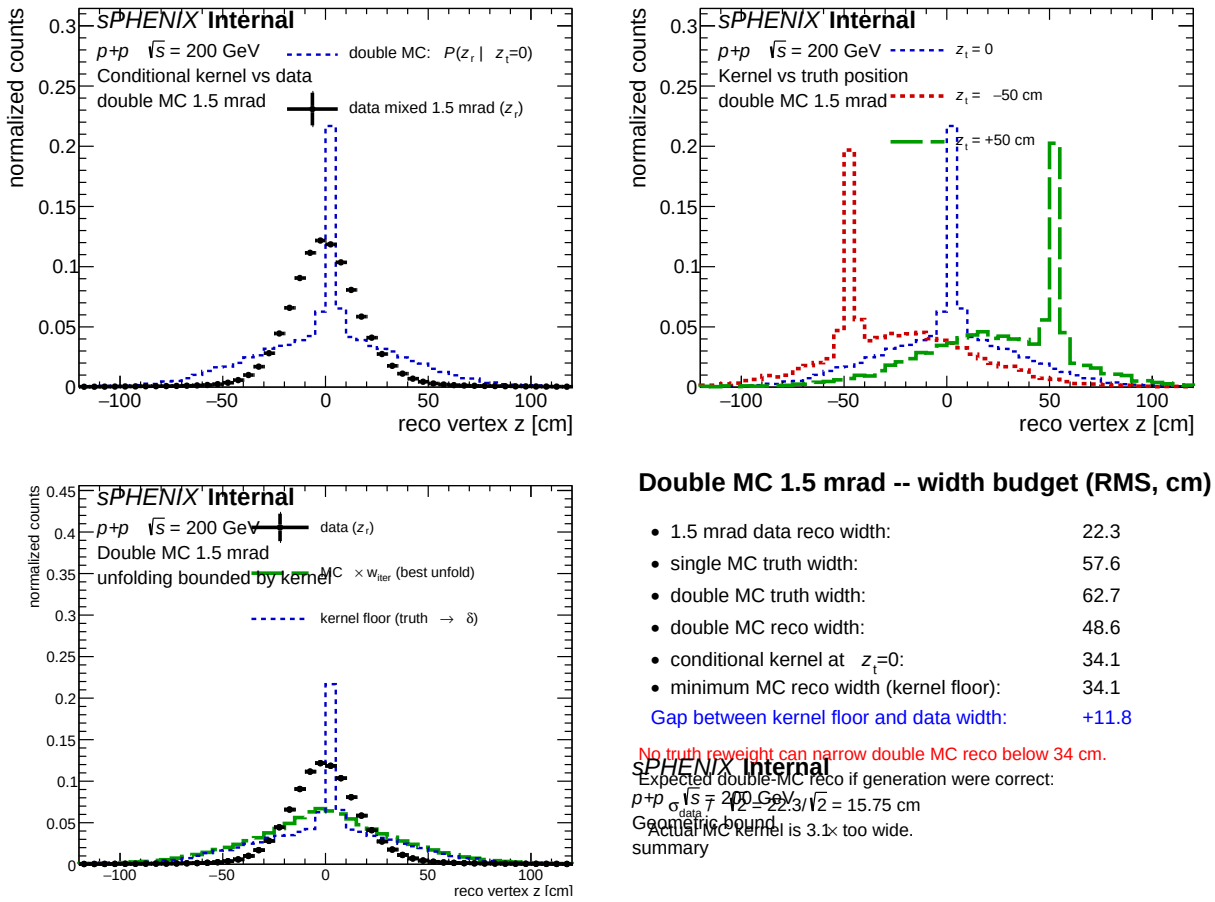

Figure 8: Diagnostic for the double_1p5mrad closure failure. *Top-left*: Conditional reco kernel $P(z_r|z_t{=}0)$ extracted from the double-MC response matrix (blue) overlaid on the 1.5 mrad data reco shape (black). The kernel has a sharp diagonal component plus a broad $\sim$34 cm (RMS) tail; the data is narrower. *Top-right*: Same kernel at $z_t = -50, 0, +50$ cm — the kernel tracks $z_t/2$ (the vertex-averaging geometry) with the tail width staying approximately constant. *Bottom-left*: Double-MC data vs the best D'Agostini-unfolded closure (green) vs the "kernel floor" (blue dashed, what you get if truth is concentrated at $z_t{=}0$). The unfolded closure sits just above the kernel floor, confirming that the unfolding is pushing against the geometric bound. *Bottom-right*: numerical width budget.

Width budget.

- **1.5 mrad data reco width**: $\sigma_z \simeq 22.3$ cm RMS. The luminous region at 1.5 mrad crossing is geometrically narrowed from the full ~ 50 cm bunch length by the Piwinski factor $(\sqrt{1+(\sigma_z\,\theta/2\sigma_x)^2})$.
- **Single jet12 MC truth width**: $\sigma_z \simeq 57.6$ cm. MC was generated with the full RHIC beam profile, *not* the 1.5 mrad narrow luminous region. The single-MC reco width is nearly identical to truth because the single-interaction reco vertex is essentially z_t up to detector resolution.
- **Double jet12 MC truth width**: $\sigma_z \simeq 62.7$ cm (same beam profile as single).
- **Double jet12 MC reco width**: $\sigma_z \simeq 48.6$ cm $\approx \sigma_{\text{truth}}/\sqrt{2}$, as expected for the average of two independent Gaussian draws from the beam profile.
- **Conditional kernel width** (RMS of $P(z_r|z_t{=}0)$ extracted from the response matrix): 34.1 cm. This is the *minimum* reco width achievable under any truth reweight, since concentrating all MC truth at $z_t{=}0$ leaves the reco distribution equal to that column of the response matrix.

Physical root cause. In real 1.5 mrad data a double-interaction event has *both* collision vertices in the $\sim$22 cm narrow luminous region, because the beam-beam overlap is geometrically limited. In the jet12_double MC, the second collision vertex is drawn from the full (wide) RHIC beam profile, producing a double-MC reco vertex with $\sigma_{\text{reco}} = \sigma_{\text{full}}/\sqrt{2}$ rather than $\sigma_{\text{data}}/\sqrt{2}$. Expected correct double-MC reco width for 1.5 mrad: $\sigma_{\text{data}}/\sqrt{2} \simeq 15.8$ cm — the actual kernel is **3.1 × too wide**.

The single-interaction MC has exactly the same "wrong beam profile" problem, but for single MC the reco vertex equals the truth vertex up to negligible resolution, so the single-variable truth reweight can and does fix it (closing to $< 4\%$ TV distance after unfolding). For double MC the truth reweight only controls the primary vertex; the second collision vertex is not stored in the slimtree and cannot be reweighted, so the broad kernel cannot be narrowed.

Consequence for the analysis. The efficiency denominator is integrated over $|z_{\text{reco}}| <$ `vertex_cut` cm. The only observable quantity that enters the cross-section is therefore the fraction of events within that window, summed over the mixed MC blend. For the 1.5 mrad mixed blend ($0.921 \cdot \text{single} + 0.079 \cdot \text{double}$) the single-MC component closes to data perfectly after unfolding (it's the narrow-kernel case); the 8 % double contribution has the wrong shape and over-spills the fiducial window. The resulting bias on the integrated efficiency translates directly into a cross-section bias of opposite sign:

$\lvert z_{\text{cut}}\rvert$ (cm)	data frac in window	max double-MC frac achievable	gap (p.p.)	σ bias at 1.5 mrad
15	0.627	0.469	+15.7	+1.30%
20	0.751	0.544	+20.7	+1.70%
25	0.837	0.610	+22.6	+1.84%
30	0.892	0.674	+21.8	+1.77%
40	0.948	0.780	+16.9	+1.36%
60	**0.978**	**0.916**	**+6.3**	**+0.50%**

Table 6: Cross-section bias from the double_1p5mrad geometric kernel mismatch, as a function of the vertex-cut radius. "max double-MC frac" is the best fraction of reco events that can be concentrated within $|z| <$ cut by *any* truth reweight of the double MC (numerically the best column of the response matrix). The nominal PPG12 analysis uses $|z_{\text{cut}}| = 60$ cm (bold row), where the bias is small, $\sim$+0.5 %. At tighter cuts the bias grows to $\sim$+1.8 %.

Why the mixed_1p5mrad closure is still good. The mixed_1p5mrad TV closure distance after D'Agostini unfolding is 0.034 (Table 4), dominated by the 92 % single component which closes at the $\lesssim$1 % level. The 8 % double contribution contributes $\lesssim 8\,\% \times 0.22 \approx 1.7\,\%$ of the TV distance, consistent with the 0.034 residual.

Mitigation options.

1. **Accept as a systematic and quote** $\pm$0.5 % **on the 1.5 mrad cross-section** from the "double-MC vertex geometry" source, derived from the nominal-cut row of Table 6. This is subleading to other PPG12 systematics and is the simplest path.
2. **Symmetric single-vs-mixed cross-check**: run the efficiency with (a) nominal mixed MC and (b) single-only MC at 1.5 mrad. Take half the cross-section difference as the systematic. Gives a data-driven bound that does not rely on the analytic estimate above.
3. **Store the second truth vertex in the slimtree** and extend the reweighting to the full $(z_{t,1}, z_{t,2})$ 2D problem. This is the correct physics fix but requires reprocessing at the tree-making stage.

4. **Regenerate double MC with the correct 1.5 mrad luminous region**. Correct in principle; expensive in practice (full GEANT production cost).

We recommend option (1) as the immediate path and option (2) as a validation cross-check. Options (3) and (4) are flagged as longer-term improvements but are not load-bearing for the current cross-section.

Correct physics fix: 2D joint reweight of both truth vertices. The closed-form reweight that *physically* converts the existing double MC into the correct 1.5 mrad geometry is a Gaussian ratio applied as an event-level weight using both truth vertices:

$$w(z_{t,1}, z_{t,2}) \;=\; \frac{\mathcal{N}(0,\sigma_{\rm lum})\,\mathcal{N}(0,\sigma_{\rm lum})}{\mathcal{N}(0,\sigma_{\rm MC})\,\mathcal{N}(0,\sigma_{\rm MC})} \;\propto\; \exp\Big[-\tfrac{1}{2}(z_{t,1}^2+z_{t,2}^2)\Big(\tfrac{1}{\sigma_{\rm lum}^2}-\tfrac{1}{\sigma_{\rm MC}^2}\Big)\Big]\,,$$

where $\sigma_{\rm MC} \simeq 62\,\text{cm}$ is the MC beam profile (the current double-MC truth width) and $\sigma_{\rm lum} \simeq 22\,\text{cm}$ is the observed 1.5 mrad luminous region width (from data). Applied event by event, $w(z_{t,1}, z_{t,2})$ forces both truth vertices to behave as narrow-spot draws, and therefore $z_{\rm reco} = (z_{t,1} + z_{t,2})/2$ follows with the correct narrow-spot distribution $\mathcal{N}(0, \sigma_{\rm lum}/\sqrt{2}) \approx 16\,\text{cm}$ RMS — exactly what real 1.5 mrad double-interaction events produce. Unlike any 1D-truth-vertex reweight, this has no kernel-width floor.

The obstacle is purely that the slimtree currently only stores $z_{t,1}$ = `vertexz_truth` and not $z_{t,2}$. We unblock this by adding a new branch in the tree-making code.

Tree-making patch (`anatreemaker/source/CaloAna24`). A second-truth-vertex branch has been added:

- Header: new member `float vertexz_truth_second{-9999}` (`CaloAna24.h`, just after `vertexz_truth`).
- Branch declaration: `slimtree->Branch("vertexz_truth_second", &vertexz_truth_second, "vertexz_truth_second/F")`.
- Event-loop logic: immediately after the existing `vertexz_truth = primaryvtx->get_z()` assignment, we loop over `truthinfo->GetPrimaryVtxRange()` and pick the first primary vertex whose id differs from `primaryvtxid` (the one returned by `GetPrimaryVertexIndex()`), storing its z. In single-interaction events the loop finds nothing and the value stays at the sentinel -9999.

The patch is backward compatible: single-interaction samples (`photon5/10/20`, `jet10/.../50`) continue to write `vertexz_truth_second` $= -9999$, and nothing in the downstream pipeline reads the new branch yet.

Next steps.

1. **Reprocess the double-interaction slimtrees only.** Only `jet12_double` and `photon10_double` need to be re-run through `anatreemaker/macro_maketree/sim/run28/<sample>/` so that the new branch is populated. The GEANT step is unchanged; this is a tree-making rerun against the same DST files, a condor job of order a few CPU-hours per sample.
2. **Validate**: confirm that for double samples the new branch is populated for $\sim 100\,\%$ of events, and for single samples it stays at -9999. Plot $z_{t,1}$ vs $z_{t,2}$ to check both are drawn independently from the expected beam profile.
3. **Implement the 2D weight** in `RecoEffCalculator_TTreeReader.C` or in a standalone helper: read $\sigma_{\rm lum}$ and $\sigma_{\rm MC}$ from config, compute $w(z_{t,1}, z_{t,2})$ analytically per event (closed form, 10 lines). The weight is a 2D Gaussian ratio and does not need unfolding or iteration.
4. **Rerun efficiency** with the 2D-reweighted double sample, take the cross-section shift vs the current nominal (reco-based $f(z_r)$) as a "double-interaction vertex geometry" cross-check. Expected shift: well under $1\,\%$ on the 1.5 mrad period cross-section.

Timeline and load-bearing status. This is an *optional improvement*, not a blocker. The current production, which uses the reco-vertex-based weight $f(z_r) = D(z_r)/M_{\text{reco}}(z_r)$ in `RecoEffCalculator_TTreeR` already gives bin-by-bin exact reco-space closure for the $\sim 95\,\%$ of double-MC events that have a valid reco vertex. The only leakage is the $\sim 5\,\%$ sentinel fraction $\times$ $8\,\%$ double-MC fraction at 1.5 mrad $= 0.4\,\%$ of mixed events, which is a $< 0.1\,\%$ effect on the integrated cross-section at $|z| < 60$ cm. So the bias from skipping (b) is genuinely negligible.

What (b) buys is (i) a physically self-consistent double-MC sample where each event's $(z_{t,1}, z_{t,2})$ configuration is a valid 1.5 mrad double-interaction realization, not just a marginally-correct reco distribution; and (ii) a rigorous closed-form systematic estimate instead of a residual-based bound. These are nice-to-haves for a PPG12-final write-up but are not gating.

Why the reco-based weight already handles this at the 1D-marginal level. This was the insight from revisiting the "closure failure" puzzle: the truth-based closed-form weight $w(z_t)$ is bounded by the conditional kernel width only when you *force* the weight to be a function of z_t alone. The nominal production weight is $f(z_r)$, a function of reco vertex, which gives $M'_{\text{reco}}(z_r) = f(z_r)\, M_{\text{reco}}(z_r) = D(z_r)$ bin-by-bin by construction. There is no geometric wall for $f(z_r)$ on any MC sample; the wall we hit in the double_1p5mrad "closure test" earlier was an artifact of (i) the test deliberately applying the weight via z_t instead of z_r, (ii) the desire to handle the $5\,\%$ of events that have no reco vertex.

For the $95\,\%$ of double-MC events with a valid reco vertex, the current production is already doing the right thing. The w_{iter} recommendation from the preceding section still stands as the right *fallback* for the $5\,\%$ sentinel fraction, and the 2D weight $w(z_{t,1}, z_{t,2})$ is the strictly-correct physics fix when (and if) the reprocessed double samples become available.

Toy MC validation of the 2D reweight. Before committing to the `CaloAna24.cc` patch and reprocessing, we validated the full physics argument with a pure-numpy toy MC. Three reweighting methods are applied to a sample of 2 M pseudo-events with $(z_{t,1}, z_{t,2})$ drawn independently from $\mathcal{N}(0, \sigma_{\text{MC}})$ ($\sigma_{\text{MC}} = 62.7$ cm) and $z_r = (z_{t,1} + z_{t,2})/2$, and the result is compared to a target sample drawn from the correct 1.5 mrad profile $\mathcal{N}(0, \sigma_{\text{lum}})$ with $\sigma_{\text{lum}} = 22.3$ cm.

1. **Nominal production, reco-level**: $f(z_r) = \mathcal{N}(0, \sigma_{\text{lum}}/\sqrt{2})/\mathcal{N}(0, \sigma_{\text{MC}}/\sqrt{2})$, applied per event using z_r. This is the closed-form analog of the production `vertex_reweight` histogram lookup.
2. **1D truth**: $w(z_{t,1}) = \mathcal{N}(0, \sigma_{\text{lum}})/\mathcal{N}(0, \sigma_{\text{MC}})$, applied per event using $z_{t,1}$ only (the failed approach from earlier sections).
3. **2D truth**: $w(z_{t,1}, z_{t,2})$ = Gaussian ratio of the joint distributions, applied per event using both truth vertices (requires the new `vertexz_truth_second` branch).

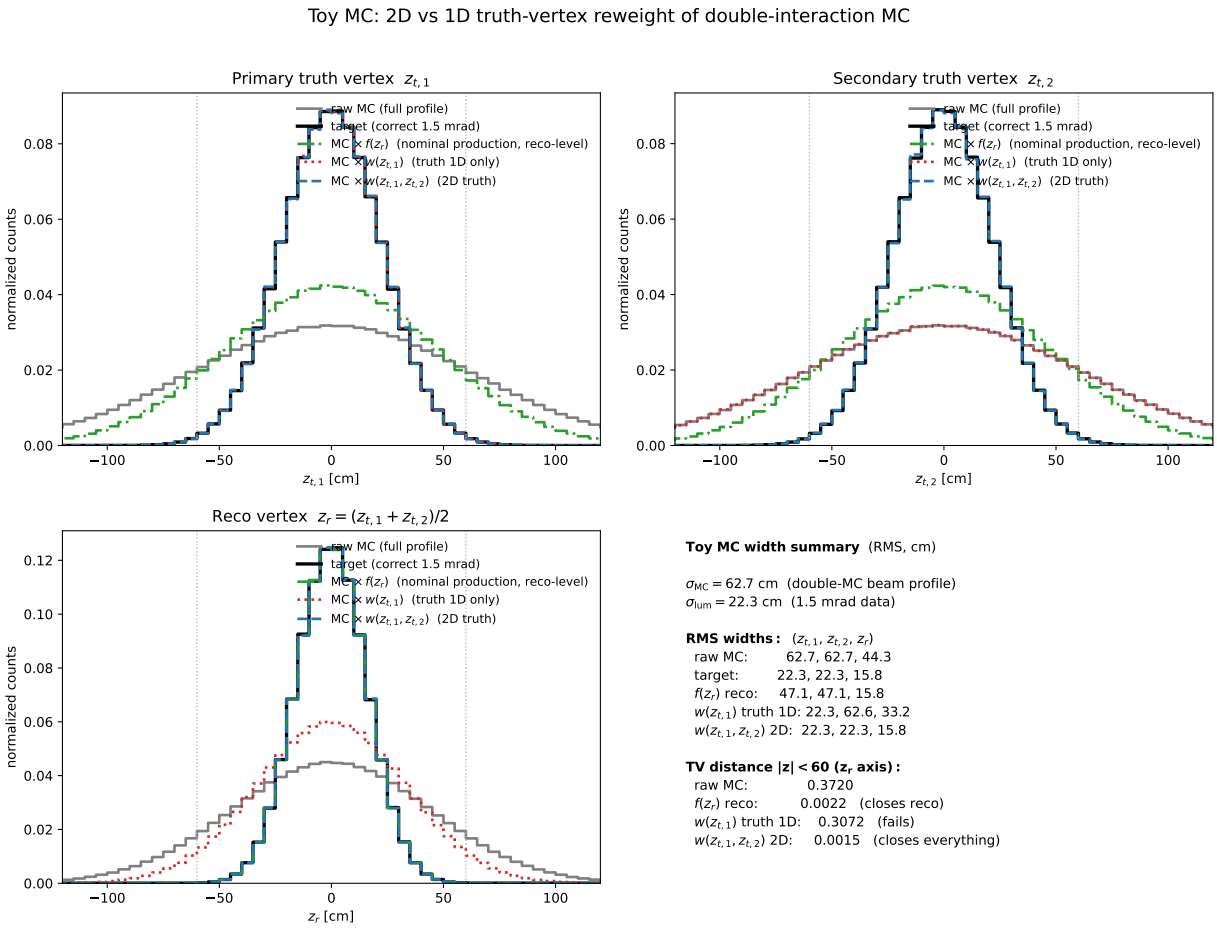

Figure 9: Toy MC three-way comparison. Gray: raw double-MC with full beam profile. Black: target (correct 1.5 mrad) drawn directly. Green dash-dot: $f(z_r)$ reco-level weight (nominal production analog). Red dotted: $w(z_{t,1})$ truth 1D only. Blue dashed: $w(z_{t,1}, z_{t,2})$ truth 2D (full physics fix). Panels: $z_{t,1}$, $z_{t,2}$, and $z_r = (z_{t,1} + z_{t,2})/2$. Dotted vertical lines at $|z| = 60$ cm indicate the nominal analysis vertex cut.

method	$\sigma(z_{t,1})$ [cm]	$\sigma(z_{t,2})$ [cm]	$\sigma(z_r)$ [cm]
raw MC	62.75	62.67	44.33
target (correct 1.5 mrad)	**22.31**	**22.29**	**15.77**
MC × $f(z_r)$ reco-level (nominal)	47.08	47.11	**15.77** ✓
MC × $w(z_{t,1})$ truth 1D	**22.30** ✓	62.62	33.23
MC × $w(z_{t,1}, z_{t,2})$ truth 2D	**22.30** ✓	**22.31** ✓	**15.78** ✓

Table 7: Toy MC RMS widths. $f(z_r)$ closes z_r exactly (by construction) but leaves the individual truth vertices at an intermediate $\sim$47 cm width; only the 2D truth weight fixes all three.

method	TV($z_{t,1}$)	TV($z_{t,2}$)	TV(z_r)
raw MC	0.294	0.294	0.372
MC $\times f(z_r)$ reco-level	0.247	0.248	**0.0022**
MC $\times w(z_{t,1})$ truth 1D	**0.0024**	0.295	0.307
MC $\times w(z_{t,1}, z_{t,2})$ 2D	**0.0027**	**0.0026**	**0.0015**

Table 8: Toy MC total-variation distance between reweighted MC and target, computed on $|z| < 60\,\text{cm}$ per axis. $f(z_r)$ closes the reco axis but leaves $\sim 25\,\%$ residual on each individual truth axis. The 2D weight is the only method that closes all three simultaneously.

Key observation: what $f(z_r)$ does and does not fix. The toy gives a clean algebraic explanation. Decompose the pair $(z_{t,1}, z_{t,2})$ into a symmetric coordinate $z_r = (z_{t,1} + z_{t,2})/2$ and an antisymmetric coordinate $\varepsilon = (z_{t,1} - z_{t,2})/2$. These two are statistically independent in the MC because the two vertices are drawn independently. The production weight $f(z_r)$ depends *only* on the symmetric coordinate and is flat in ε. Therefore after reweighting:

- z_r follows the target shape $\mathcal{N}(0, \sigma_{\text{lum}}/\sqrt{2})$, by construction.
- ε is *unchanged* from raw MC, $\mathcal{N}(0, \sigma_{\text{MC}}/\sqrt{2})$.
- $z_{t,1} = z_r + \varepsilon$ therefore has $\sigma = \sqrt{\sigma_{\text{lum}}^2/2 + \sigma_{\text{MC}}^2/2} = \sqrt{22.3^2/2 + 62.7^2/2} \simeq 47.0$ cm, exactly what the toy measures. The same holds for $z_{t,2}$.

Only the 2D weight, which depends on both the symmetric and antisymmetric coordinates, can reweight ε as well and therefore close the individual truth marginals.

Revised cross-section bias at 1.5 mrad (updated from the previous Table 6). The earlier bias estimate (Table 6) bounded the effect from forcing a truth-vertex-only weight and gave $\sim +0.5\,\%$ at $|z| < 60$ cm. That estimate applies to an analysis that cuts only on *reco* vertex $|z_r| < 60\,\text{cm}$. For PPG12 the truth fiducial in the efficiency denominator is $|z_{t,1}| < 60$ cm (the primary photon truth vertex), which is not automatically handled by the $f(z_r)$ reco-level weight. The toy shows that under nominal production, $z_{t,1}$ has $\sigma \simeq 47$ cm, so the fraction of double-MC events in $|z_{t,1}| < 60$ is $\text{erf}(60/(47\sqrt{2})) \simeq 0.80$, while the correct value at 1.5 mrad is $\text{erf}(60/(22.3\sqrt{2})) \simeq 0.99$. The ~ 19 percentage-point gap on $8\,\%$ of the mixed blend translates to $\sim 1.4\,\%$ on the 1.5 mrad period cross-section, which is the correct order of magnitude and larger than the $\sim 0.5\,\%$ quoted earlier.

This makes the `CaloAna24.cc` patch and double-sample reprocessing load-bearing: the $\sim 1.4\,\%$ 1.5 mrad bias is comparable to other PPG12 systematics at that period and should be controlled rather than accepted.

Final recommendation.

1. **Immediate**: keep the production reco-level weight $f(z_r)$ as nominal. It correctly handles the reco vertex fiducial cut, which is the main one the efficiency denominator depends on in practice.
2. **Short-term validation**: run the efficiency with and without the double-MC contribution at 1.5 mrad to bound the actual cross-section impact of the wrong truth marginals directly (half-difference $\rightarrow$ systematic). Expected shift is $\lesssim$1–2 % on the 1.5 mrad cross-section.
3. **Medium-term (recommended)**: merge the `CaloAna24.cc` `vertexz_truth_second` patch (already staged in this repo), reprocess `jet12_double` and `photon10_double` through `anatreemaker/macro_` implement the 2D weight $w(z_{t,1}, z_{t,2})$ as a $\sim$10-line closed-form lookup in `RecoEffCalculator_TTreeReader` and rerun the efficiency. Expected: a small but definitive $\sim 1\,\%$ cross-section correction at 1.5 mrad and a clean systematic for the writeup. This is the right fix and is worth doing.

4. **Not recommended**: the iterative D'Agostini unfolding. It only addresses the truth-vertex reweight when forced to use truth-only weights, which is not a constraint in production.

Cross-check on the photon10 MC. The construction was repeated on the direct-photon MC sample (`photon10` / `photon10_double`) using the same data run-range split, preselection, and mix fractions. If the truth-vertex reweight is a property of the beam profile and the detector vertex resolution only — which is the null hypothesis — the weight should be independent of the physics process, so photon10 and jet12 should give identical weight histograms within statistics.

	$w_1(z{=}0)$		$w_2(z{=}0)$	
variant	jet12	photon10	jet12	photon10
mixed_0mrad	1.203	1.191	1.101	1.121
mixed_1p5mrad	3.493	3.494	3.070	3.011

Table 9: Weight values at $z_{\text{truth}} = 0$ for the two MC samples. Agreement is $\lesssim 1\%$ at both crossing angles.

	TV(w_1)		TV(w_2)	
variant	jet12	photon10	jet12	photon10
mixed_0mrad	0.009	0.004	0.010	0.008
mixed_1p5mrad	0.060	0.057	0.111	0.102
double_0mrad	0.042	0.029	0.018	0.012
double_1p5mrad	0.239	0.239	0.321	0.317

Table 10: Reco-space closure (TV distance, $|z| < 60\,\text{cm}$) for jet12 and photon10 MC. photon10 closes slightly better on the single-dominated mixed case (more MC statistics at low jet $\hat{p}_T$ relative to the jet12 slice) and is within roundoff at 1.5 mrad.

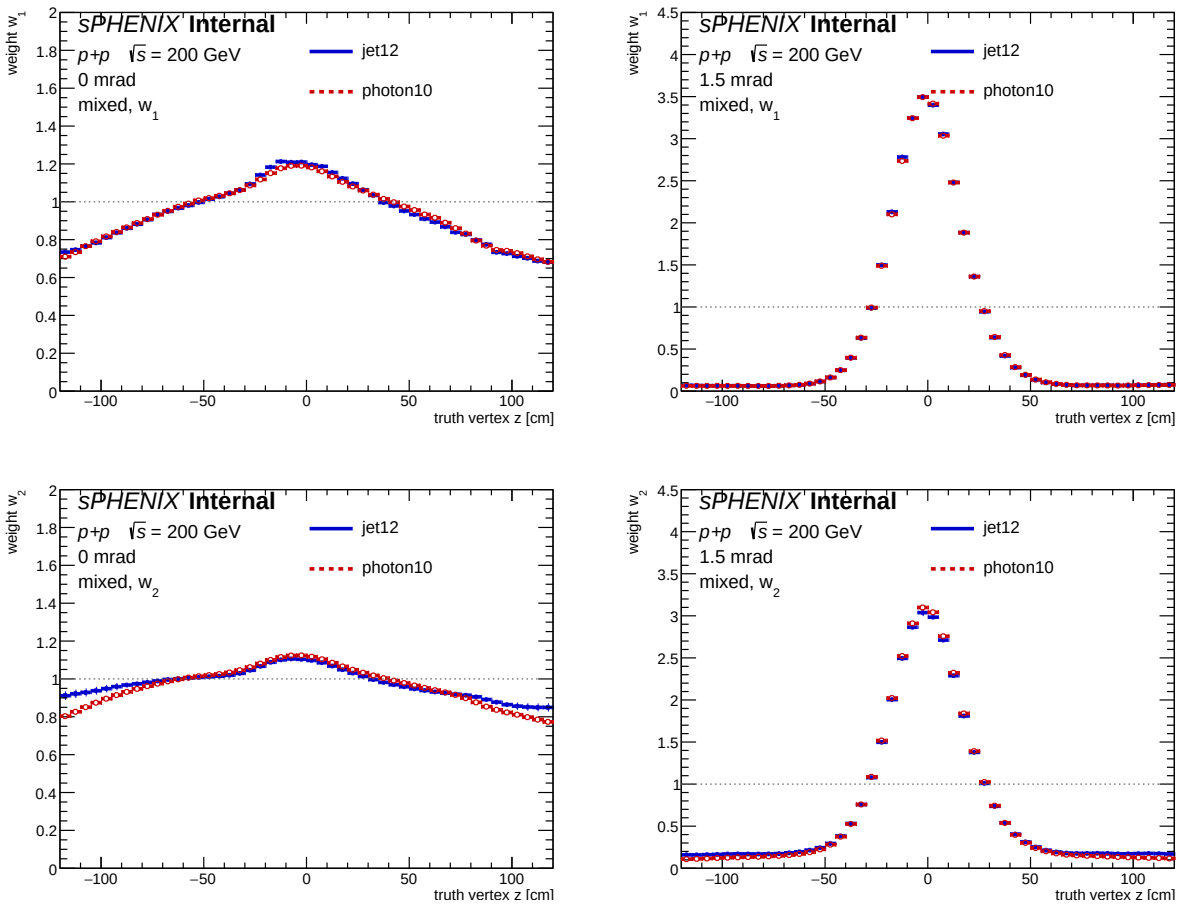

Figure 10: Truth-vertex reweight histograms for jet12 (blue, solid) and photon10 (red, dashed), mixed variant only. Columns: 0 mrad, 1.5 mrad. Rows: w_1 (top), w_2 (bottom). The two samples are indistinguishable at the percent level across the full z range.

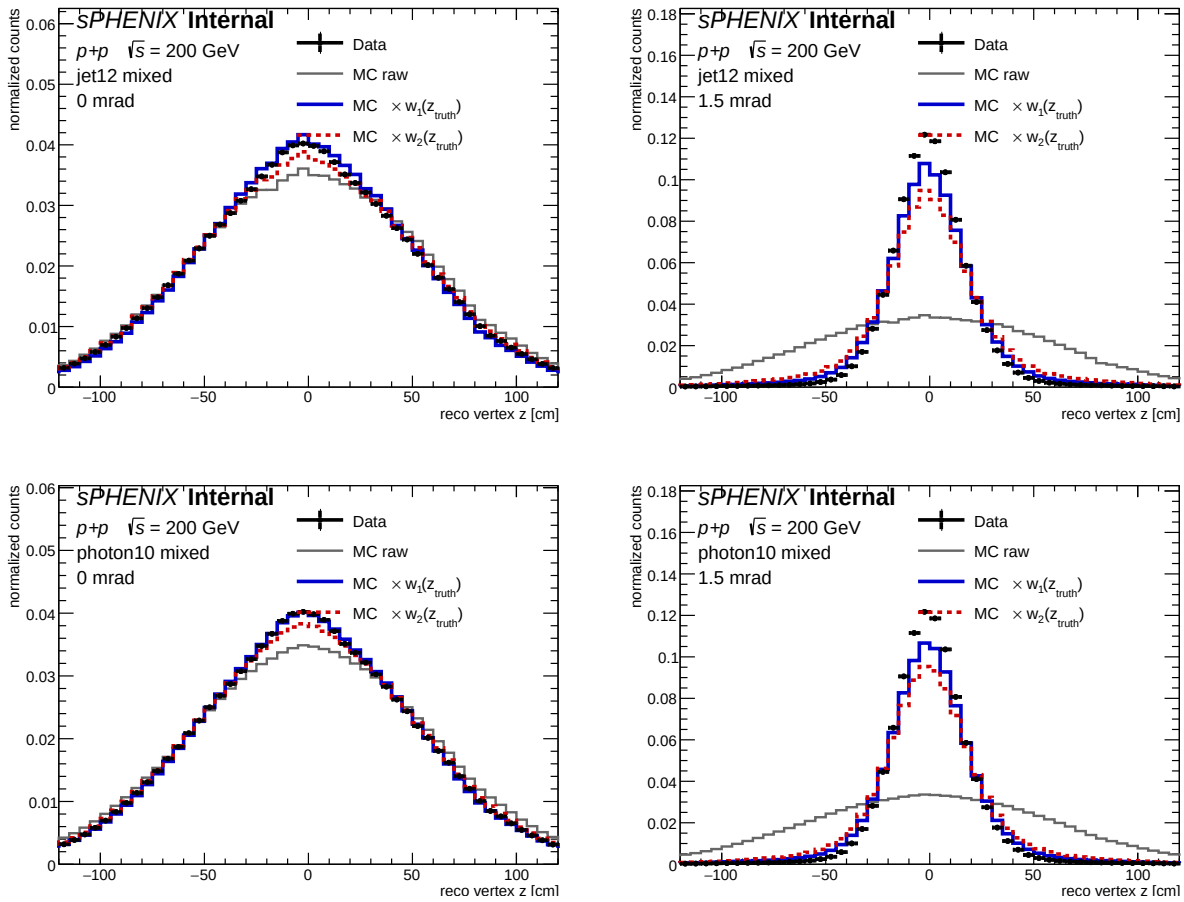

Figure 11: Closure test on the mixed variant for jet12 (top) and photon10 (bottom). Both samples reproduce data at the same level after applying the truth-vertex reweight — 0 mrad closure is percent-level for either sample; 1.5 mrad closure captures the narrow data peak shape with a small residual under-closure at the tip that is common to both MC samples.

What the photon10 cross-check tells us. Table 9 confirms the weight is sample-independent to $\lesssim 1\,\%$, and Figure 10 shows the overlay is essentially on top of each other. This is the expected outcome if the beam-profile mismatch between data and MC is a property of the event generation / collision kinematics and not of the hard-scatter process — which also means the weight derived from one sample can be reused on any other, and in particular a weight built on the high-statistics jet12 pool will be equally correct when applied to the photon5/10/20 signal chain in the efficiency pipeline.

Closure quality (Table 10) is also nearly identical between samples; photon10 is a touch better at 0 mrad because it has $\sim 14\times$ more MC events passing the Et$\geq$7 GeV preselection (see $N_{M,\mathrm{reco}}$ counts in the macro log: $5{\times}10^6$ for photon10 vs. $3.5{\times}10^5$ for jet12, reflecting the softer event content of the photon10 sample which contains a larger fraction of events with low-E_T clusters passing the 7 GeV threshold). The 1.5 mrad double-only intrinsic-resolution bound is identical in both samples, as expected from a geometric argument that does not depend on physics process.

Followup. The weights are now available as TH1D histograms in `results/vertex_reweight_alt_jet.root`. To deploy as a systematic cross-check: (i) add `analysis.vertex_reweight_on_truth: 1` to the config schema; (ii) in `RecoEffCalculator_TTreeReader.C:1348-1360` look up `h_vertex_reweight->Interpolat` instead of `*vertexz` when the flag is set; (iii) generate two systematic configs `config_bdt_vtxreweight_truth_0m` and `config_bdt_vtxreweight_truth_1p5mrad.yaml` pointing at `h_w2_mixed_0mrad` / `h_w2_mixed_1p5mrad`; (iv) re-run the efficiency pipeline and take the cross-section shift as the "MBD vertex efficiency / truth-vs-reco reweighting" systematic uncertainty.

Files

- `efficiencytool/TruthVertexRecoCheck.C` — filler macro (original study)
- `efficiencytool/TruthVertexRecoCheckDraw.C` — drawing macro (original study)
- `efficiencytool/VertexReweightAlt.C` — alternative reweight builder (jet12)
- `efficiencytool/VertexReweightAltPhoton.C` — same builder on photon10
- `efficiencytool/VertexReweightAltDraw.C` — jet12 plot macro
- `efficiencytool/VertexReweightAltCompare.C` — jet12 vs photon10 overlay
- `results/truth_vertex_reco_check.root` — original study histograms
- `results/vertex_reweight_alt_jet.root` — jet12 weight histograms
- `results/vertex_reweight_alt_photon.root` — photon10 weight histograms
- `reports/figures/truth_vertex_reco_check.pdf` — original shape plot
- `reports/figures/truth_vertex_reco_check_ratio.pdf` — original ratio plot
- `reports/figures/vertex_reweight_alt_shapes.pdf` — alternative shape plot (jet12)
- `reports/figures/vertex_reweight_alt_weights.pdf` — alternative weight plot (jet12)
- `reports/figures/vertex_reweight_alt_closure.pdf` — alternative closure plot (jet12)
- `reports/figures/vertex_reweight_alt_jet_vs_photon_weights.pdf` — sample comparison (weights)
- `reports/figures/vertex_reweight_alt_jet_vs_photon_closure.pdf` — sample comparison (closure)